Titanic Survival Prediction Report

Objective

The objective of this project is to predict passenger survival on the Titanic using machine learning models.

Dataset Overview

The Titanic dataset contains demographic and travel-related information for passengers aboard the Titanic.

Important attributes include:

- Passenger Class
- Gender
- Age
- Fare
- Number of Family Members
- Embarkation Port

Target Variable:

- Survived

Exploratory Data Analysis

The dataset was explored using:

- shape()
- info()
- describe()

This helped identify missing values and understand feature distributions.

Data Cleaning

The following preprocessing steps were performed:

1. The deck column was removed because it contained a large number of missing values.
2. Rows containing missing values were removed using dropna().

Feature Engineering

A new feature named FamilySize was created using:

FamilySize = SibSp + Parch + 1

This feature represents the total number of family members travelling with a passenger, including the passenger.

Data Transformation

Categorical features were converted into numerical values:

Sex:

- Male = 0
- Female = 1

Embarked:

- S = 0
- C = 1
- Q = 2

Model Training

Two machine learning models were trained:

1. Logistic Regression
2. Random Forest Classifier

The dataset was split into:

- 80% Training Data
- 20% Testing Data

using train_test_split() with random_state = 42.

Evaluation Results

Logistic Regression Accuracy: 79.02%

Random Forest Accuracy: 77.62%

Model Comparison

Logistic Regression achieved slightly higher accuracy than Random Forest on the test dataset.

Although Random Forest can model more complex relationships, Logistic Regression performed better for the selected features and preprocessing approach.

Deployment Decision

Logistic Regression was selected for deployment because it achieved the highest test accuracy.

Advantages:

• Higher predictive performance
• Faster training
• Lower computational cost
• Easier interpretation

Conclusion

The Titanic survival prediction project successfully demonstrated a complete machine learning workflow including data preprocessing, feature engineering, model training, evaluation, and model selection.

Among the evaluated models, Logistic Regression achieved the best performance and was selected as the final model.